

Course code: ECA3102 Course: Nano electronics Exp 1: Simulation of Density of States of Quantum well structure with various nano meter thickness.

Why ? The purpose of this project is to understand how **quantum confinement** changes the **Density of States (DOS)** when the thickness of a quantum well is varied from a few nanometers to several nanometers.

The **Density of States** tells us

How many electron energy states are available at each energy level.

for Nanocrystalline Composites

Basic Model Configuration

Advanced Model Configuration

Temperature: ☐ 300K

Temperature Gradient: ☐ 10K

Silicon Doping:

Germanium Doping:

Substrate Composition Si(x)Ge(1-x):

Mat 1 Material:

Mat 2 Material:

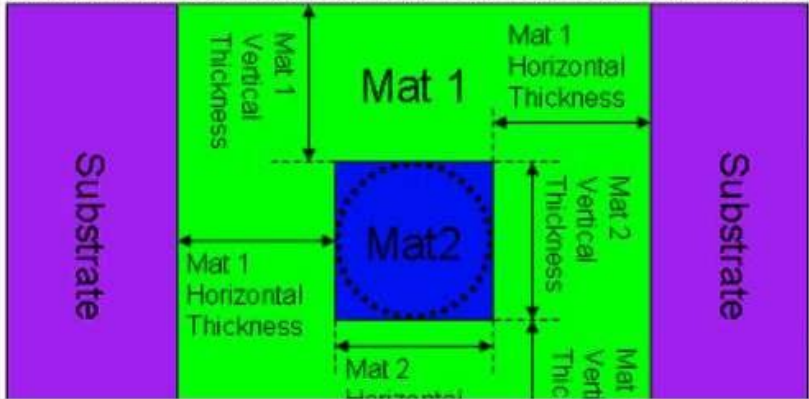
Mat 1 Horizontal Thickness:

Mat 1 Vertical Thickness:

Mat 2 Horizontal Thickness:

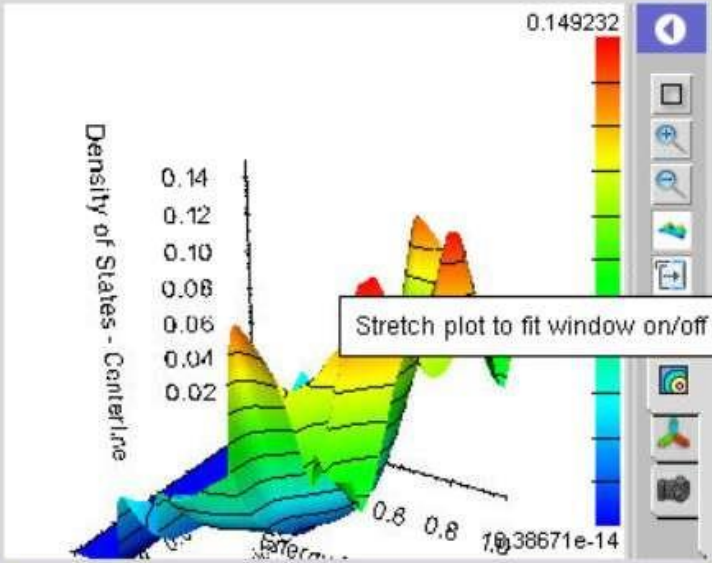
Mat 2 Vertical Thickness:

Reflective Boundary Condition



Simulate

Result:



2 results Clear One Clear All

All

Simulation = #2

Mat 1 Horizontal Thickness = 0.2nm

Mat 1 Vertical Thickness = 0.2nm

Mat 2 Horizontal Thickness = 0.2nm

Mat 2 Vertical Thickness = 0.2nm

barrierLx = 0.2nm

barrierLy = 0.2nm

wellLx = 0.2nm

wellLy = 0.2nm

Storage (manage)

0% of 10.000004GB

780 × 600

Case study 1: Simulate the density of states of quantum well of thickness 0.5 nm
Aim: To simulate the **Density of States (DOS)** of a quantum well structure with a **0.5 nm well thickness** and analyze the effect of strong quantum confinement on the available electron energy states.

Thermoelectric Power Factor Calculator for Nanocrystalline Composites

Basic Model Configuration

Advanced Model Configuration

Temperature: 300K

Temperature Gradient: 10K

Silicon Doping: 1e+18/cm3

Germanium Doping: 1e+18/cm3

Substrate Composition Si(x)Ge(1-x): 1

Mat 1 Material: Silicon

Mat 2 Material: Germanium

Mat 1 Horizontal Thickness: 0.5nm

Mat 1 Vertical Thickness: 0.5nm

Mat 2 Horizontal Thickness: 0.5nm

Mat 2 Vertical Thickness: 0.5nm

Reflective Boundary Condition

Simulate

Result: Density of States - Centerline

2 results

Clear One

Clear All

All

Simulation = #2

Mat 1 Horizontal Thickness = 0.5nm

Mat 1 Vertical Thickness = 0.5nm

Mat 2 Horizontal Thickness = 0.5nm

Mat 2 Vertical Thickness = 0.5nm

barrierLx = 0.5nm

barrierLy = 0.5nm

wellLx = 0.5nm

wellLy = 0.5nm

Case study 2: Simulate the density of states of quantum well of thickness 0.6 nm

Aim: To simulate the **Density of States (DOS)** of a quantum well structure with a **0.6 nm well thickness** and investigate the effect of increased well thickness on the electronic energy states.

for Nanocrystalline Composites

Basic Model Configuration

Advanced Model Configuration

Temperature:

Temperature Gradient:

Silicon Doping:

Germanium Doping:

Substrate Composition Si(x)Ge(1-x):

Mat 1 Material:

Mat 2 Material:

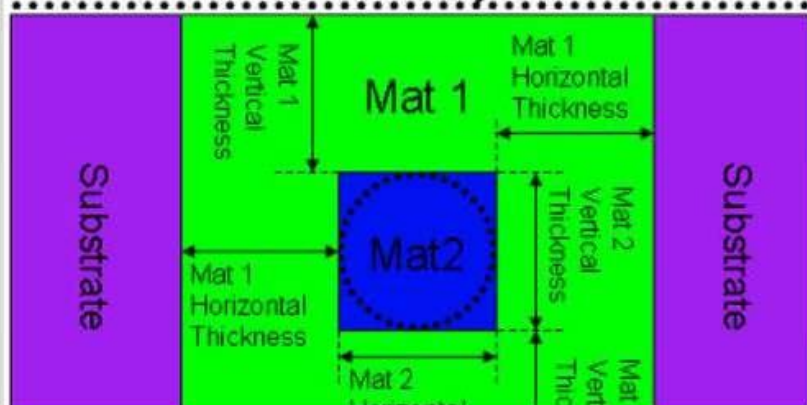
Mat 1 Horizontal Thickness:

Mat 1 Vertical Thickness:

Mat 2 Horizontal Thickness:

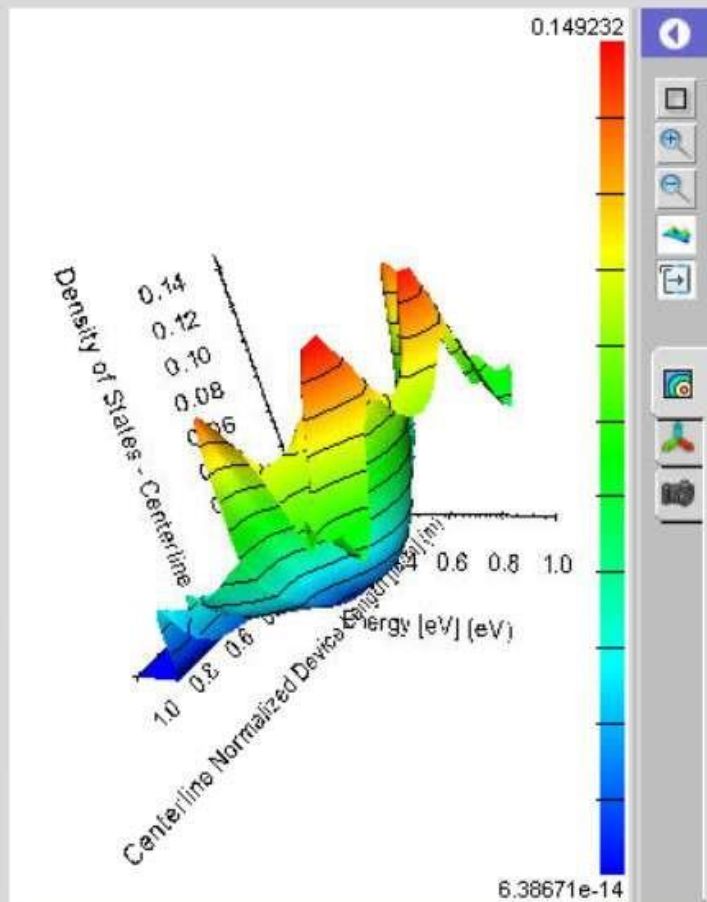
Mat 2 Vertical Thickness:

Reflective Boundary Condition



Simulate

Result:



1 result

Clear

Storage (manage)

0% of 10.000004GB



780 × 600